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Si5351A Breakout Board TCXO Upgrade – Ripples in the Ether

~3 minutes

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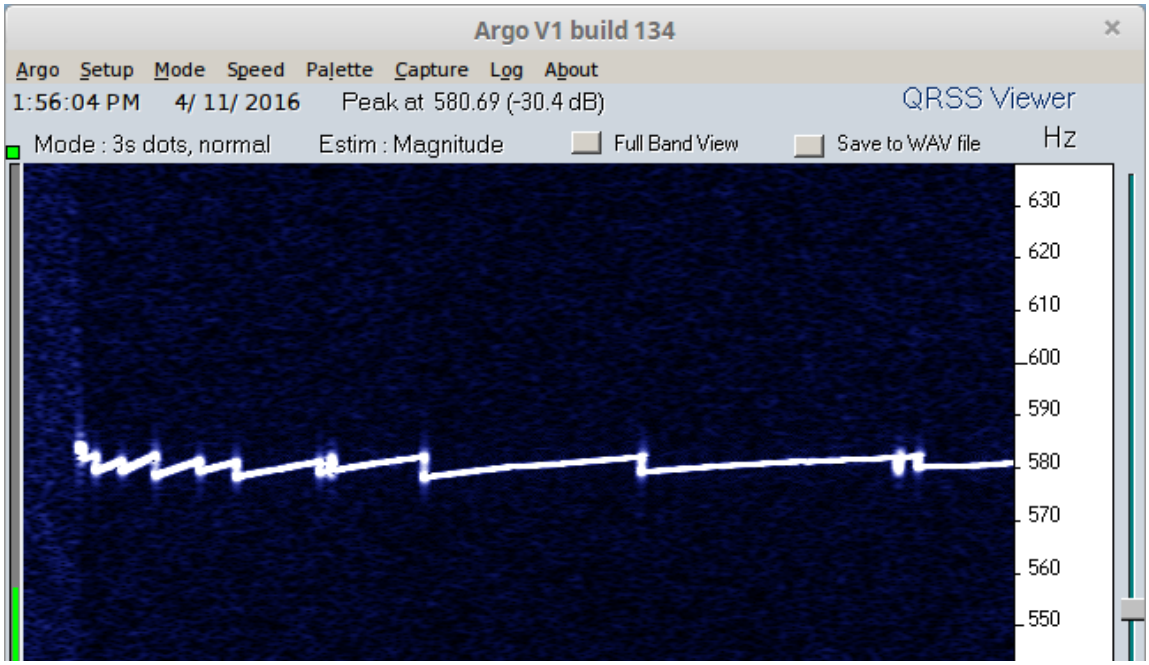
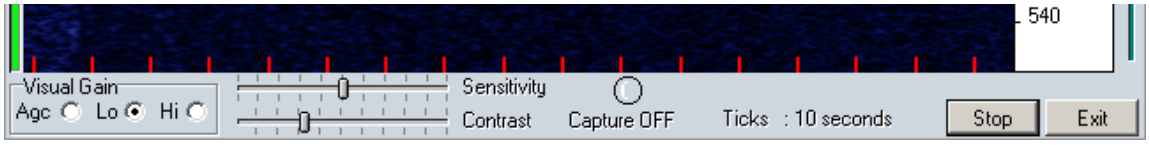
I'm pleased to announce an upgrade to the Etherkit Si5351A Breakout Board with TCXO reference oscillator. In boards manufactured previous to today (and the ones kitted in the initial crowdfunded initiative), the TCXO used was a Fox FOX924B-25.000. It worked well and did the job it was supposed to, but has one flaw in certain situations. The compensation loop in this particular TCXO has relatively wide frequency adjustment steps that can become obvious when using the Si5351A Breakout Board in a very narrow band mode such as QRSS. Here is a screen capture from Argo showing the behavior of the Si5351A Breakout Board with the Fox TCXO when outputting a carrier on 28 MHz, from a cold start.

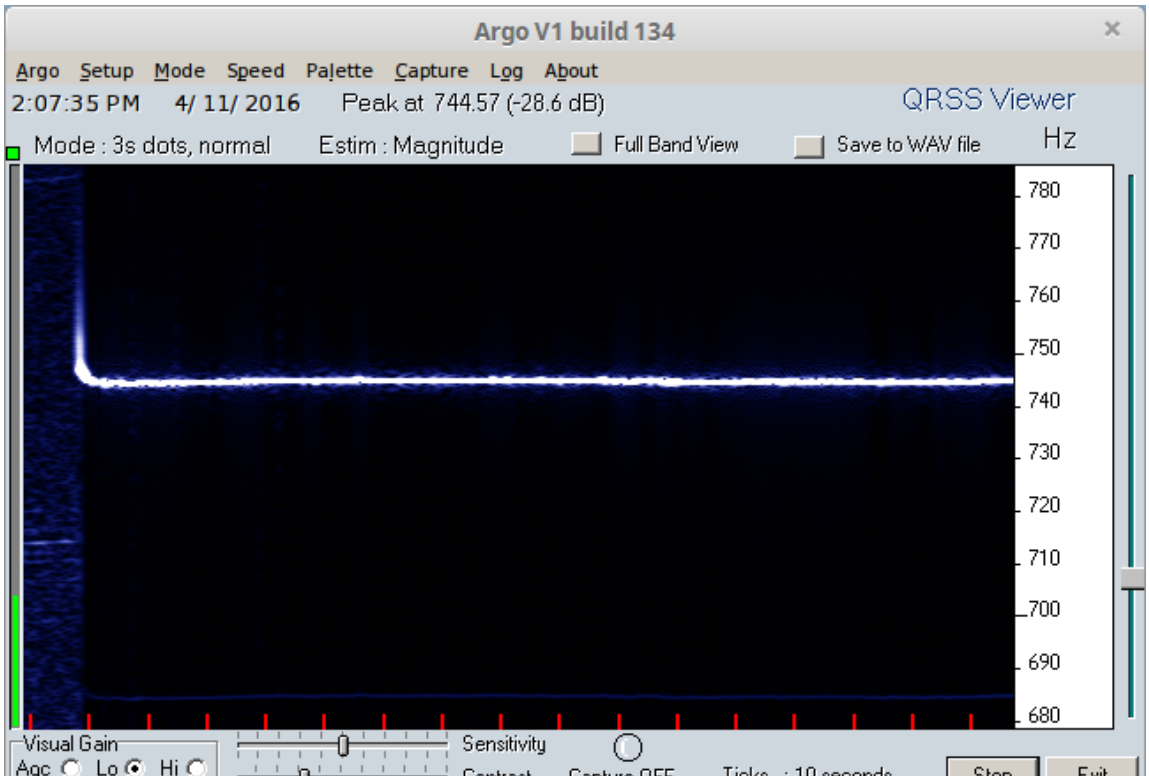




Fox FOX925B-25.000

As you can see, as the TCXO is stabilizing to operating temperature, the compensation circuit adjusts the frequency in relative course steps of around 3 Hz or so. Once the TCXO is warmed up, it jumps around less often, but still does occasionally have to frequency correct, and does so in a jump of similar size. This TCXO is still fine for most other uses, such as a VFO for a SSB/CW radio, but doesn't work so great for the MEPT modes, especially on the upper frequency range of the Si5351.

In the course of developing OpenBeacon 2, I determined that I would need to find a more suitable TCXO to use with the Si5351. After trying about five different oscillators, I finally found a fantastic substitute, and it only costs a bit more than the Fox TCXO. The Abracon ASTX-H11-25.000MHZ-T comes in a smaller package than the Fox FOX924B-25.000, but it will still fit on the footprint for X1 on the Si5351A Breakout Board. Here you can see the same test as above run for the Abracon TCXO.



Abracon ASTX-H11-25.000M

It seems to find a stable frequency very quickly, and more importantly you can see that the compensation loop seems much tighter, with frequency corrections coming much faster and in smaller increments. You can only barely see a bit of fuzz from the frequency corrections on this low-bandwidth plot. This is much, much better for the MEPT modes.

As of today, all of the Si5351A Breakout Boards with TCXO option sold in the Etherkit store will have the Abracon TCXO, and the price will stay the same at $15.

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